# Supervised Bayesian Matrix Factorization Identifies Prognostic Gene Expression Programs

Presentation Notes

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This companion is a preparation guide for an approximately 15–20 minute lab meeting talk. This deck is built directly from the 6/18 template but is **not a diff against it** — it presents the current state of the method, not a changelog. Two changes run through the entire talk: the loadings predictor ($L\beta$) is dropped from the narrative entirely (only the projection predictor $(YF)\beta$ is presented), and $K$ selection is a **two-stage split** — a consensus of ELBO, BIC, log-likelihood, and cross-validated external C-index picks $K_{\text{init}}$; ARD shrinkage then determines $K_{\text{eff}}$ from that one fit, with no separate per-K validation loop for $K_{\text{eff}}$ itself. Cross-validation is *not* gone; it is one of four inputs to $K_{\text{init}}$, not the sole method, and it no longer determines $K_{\text{eff}}$ on its own.

## Executive Summary

Pancreatic ductal adenocarcinoma (PDAC) needs validated molecular prognosis, but the conventional *two-step* pipeline — unsupervised factorization, then test factors for survival after the fact — chases the largest expression variance and misses low-variance prognostic programs. This work is a *joint, supervised Bayesian matrix factorization*: the projection predictor $\eta = (YF)\beta$ scores new patients with the identical formula used at training, so external scoring is exact and leakage-free.

- **$K$ selection, reframed:** $K_{\text{init}}$ is chosen from a consensus of ELBO, BIC, an ELBO-style joint log-likelihood, and cross-validated external C-index, swept over $K_{\text{init}} = 2..20$. The four criteria *disagree* — ELBO/BIC prefer $K_{\text{init}} = 3$, external C-index prefers $K_{\text{init}} = 12$ — and that disagreement is presented plainly, with $K_{\text{init}} = 7$ kept as the practical recommendation (stable $K_{\text{eff}} = 2$, externally competitive, reachable from a single fit).
- **Findings:** On real PDAC data (train TCGA-PAAD + CPTAC, validate five held-out cohorts) the model attains mean external C = **0.627**, beating the fairest independent two-step baseline (EBMF → Cox, $K = 40$, LASSO stage 2) by a pooled bootstrap **+0.026** (95% CI [0.0002, 0.0498], significant). A signal-strength sweep shows the joint model matches the unsupervised baseline when there is no survival signal to exploit, and pulls ahead as signal grows.
- **Simulation, re-run under ARD:** the multi-cohort simulation now selects $K$ the way real data does (over-specified $K_{\text{init}}$ + ARD pruning) instead of an oracle true-$K$. Recovery, specificity, and C-index are flat across $K_{\text{init}}$ (15 seeds) — over-specifying $K$ costs nothing. A secondary false-positive-rate question trends favorably but is **not statistically significant** ($p = 0.07$) — reported as a trend, not a finding.

- **Parsimony:** of the programs $K_{\text{init}} = 7$ keeps, only 2 carry survival signal — an adverse basal-like program and a protective classical program, both now biologically annotated by 5 independent methods; the 2 genomics-only kept programs are newly characterized too (stroma, immune infiltrate).

## Opening

*Slide numbers below match the deck's on-screen numbering (title slide = 1). The three section-divider slides (3 Background, 5 Methods, 12 Results) count toward that numbering; appendix slides are "uncounted" in the deck and are labeled A1–A4 here.*

**Pacing.** 15–20 minutes over 24 slides is under one minute per slide on average — the appendix (A1–A4) is Q&A-only, so budget the pace against the 24 counted slides, not 28. Text in **navy bold** is a must-say point; **underlined Carolina blue** marks headline numbers.

### Slide 1 — Title Slide

**Script.** Thanks everyone — a project update on the supervised Bayesian matrix factorization work. Since June, two things changed: **$K$ selection no longer relies on cross-validation alone** — it is now a consensus of fit criteria plus ARD shrinkage — and **only the projection predictor is presented going forward**. Frame the talk as a methods update and a validated results update.

- **Main Takeaway:** This deck describes the current state of the method, not a list of changes since June — avoid "here's what's new" framing throughout.
- **Timing:** Two sentences, then the agenda.

## Main Deck

### Slide 2 — Agenda

**Script.** A couple of minutes on motivation, then the method — most of it is the two-stage $K$-selection framework — then **most of the talk is results**: the $K_{\text{init}}$ sweep, gene programs and survival, the joint-vs-two-step comparison, and the multi-cohort simulation. Close with impact and next steps.

- **Main Takeaway:** Results-forward talk; the K-selection story is the methodological thread running through everything.

### Slide 3 — Section divider: Background

**Script.** (Divider — no content.) Pause to transition from agenda into motivation.

### Slide 4 — Background: PDAC needs molecular prognosis

**Script.** PDAC has five-year survival under 12% and no adopted molecular stratification at diagnosis. Goal: recover **gene expression programs** — a column of $F$ with patient activation in $L$ — that predict survival, validated on independent cohorts. Why jointly rather than two-step: unsupervised axes chase the largest variance, often batch; **a 15%-variance program may be batch while a 3% one separates survivors**. Supervision steers factors toward survival-relevant structure during estimation.

- **Main Takeaway:** Genuine clinical need; joint-over-two-step is the thread of the whole talk.
- **Good Line to Stress:** "A high-variance program can be batch effect; a low-variance program can be the one that separates survivors. Supervision prioritizes the second."

### Slide 5 — Section divider: Methods

**Script.** (Divider — no content.) Signal the shift into method.

### Slide 6 — The model — two coupled components

**Script.** Two coupled pieces: $Y = LF^\top + E$ for expression, $h_i(t) = h_0(t)\exp(\eta_i)$ for survival, coupled through the **shared gene-program weights $F$**. The table defines every symbol, including $\eta_i = (Y_i F)\beta$ — the projection predictor is the only one shown now. A program is **prognostic only if $\beta_k \neq 0$**. Baseline hazard $h_0(t)$ is non-parametric — cancels in the partial likelihood, no shape assumption.

- **Main Takeaway:** Shared $F$ drives both genomics reconstruction and the survival score; only the projection predictor is discussed from here on.
- **Good Line to Stress:** "$\beta_k = 0$ means the model keeps that program for expression but not for prognosis — that separation is the whole point."

### Slide 7 — The model: projection predictor and explicit priors

**Script.** The projection predictor $\eta = (YF)\beta$ scores a new patient with the **identical training formula** — exact, leakage-free, no OLS re-projection approximation. Priors, stated explicitly: **point-exponential (non-negative) on $L, F$, normal on $\beta$**, closed-form MLE for $\tau$ — **not point-normal**. The normal prior on $\beta$ is what makes **ARD-based $K$ selection possible** two slides from now — shrinkage toward zero, not a hard spike-and-slab switch.

- **Main Takeaway:** Exact external scoring is why this is the recommended (and now only) linear predictor.
- **Emphasis:** Flag the normal-on-$\beta$ detail explicitly — it is the mechanism the $K$-selection slide relies on.
- **Likely Question:** "Why drop $L\beta$ entirely — wasn't it the fully-joint reference?" It still exists in the codebase, but the projection model has always been the recommendation (exact scoring, no re-projection mismatch), and simplifying the narrative to one predictor keeps focus on the results that matter for this update.

### Slide 8 — The objective — ELBO & variational family

**Script.** Mean-field variational family; CAVI maximizes the **ELBO** — Gaussian genomics fit, a 2nd-order Taylor Cox survival term (keeps it conjugate), and a per-block KL term (the adaptive shrinkage). Shared $F$ in both likelihood terms *is* the supervision. **No $\alpha$ to tune** ($\lambda = 1$). New since June: the ELBO is now **also one of the four $K_{\text{init}}$-selection criteria**, not excluded from it.

- **Main Takeaway:** Supervision is one joint objective; the ELBO now does double duty as fitting objective *and* one $K_{\text{init}}$-selection input.
- **Likely Question:** "Doesn't using the ELBO to fit *and* to help pick $K$ double-count?" No — fitting happens at a fixed $K_{\text{init}}$; comparing ELBO *across* different $K_{\text{init}}$ fits afterward is a separate, standard model-comparison use of the same quantity, analogous to comparing log-likelihoods across models with different parameter counts (which is exactly why BIC, which penalizes for that, is shown alongside it).

### Slide 9 — Inference — factor-wise CAVI

**Script.** Coordinate-ascent, one factor at a time, each update an EBNM sub-problem. $\beta_k$'s precision is governed by its own projection score's variance, so a **no-signal factor shrinks $\beta_k \to 0$ on its own** — the mechanism ARD relies on. Convergence: relative ELBO change below $10^{-5}$ ($\Delta L, \Delta\beta$ are diagnostics only). Sign correction: orient by **training** concordance, reuse at test.

- **Main Takeaway:** One factor at a time; convergence is judged on the ELBO itself.
- **Good Line to Stress:** "A factor with no survival signal shrinks its own $\beta$ toward zero as a natural consequence of the update, not because of an external rule — that's ARD."

### Slide 10 — Preprocessing is crucial — per-platform normalization

**Script.** Unchanged from June, still the most important practical finding. $A_k = \sum_i w_i (YF)^2_{ik}$; without per-platform z-standardization, between-platform variance swamps $A_k$ and $\beta \to 0$. **10 of 12** non-per-platform pipelines collapse in an 18-config benchmark. No leakage — normalization fit within each training cohort before any split.

- **Main Takeaway:** Per-platform normalization before merging is the difference between a working model and $\beta \to 0$.
- **Likely Question:** "Isn't per-cohort normalization leakage?" No — fit within each training cohort before any split; held-out cohorts get the same per-cohort recipe at scoring time, never using training or test labels.

### Slide 11 — Selecting the number of programs: a two-stage framework

**Script.** The core methods update. **Stage 1** picks $K_{\text{init}}$ from a consensus of four criteria: ELBO, BIC (df $= K_{\text{init}} \cdot (n + p + 1)$, charged by starting $K$, not $K_{\text{eff}}$), an ELBO-style log-likelihood, and cross-validated external C-index — **these need not agree**, and the full curve is shown in Results, not just the answer. **Stage 2** is ARD: at the chosen $K_{\text{init}}$, ‘classify_factors()‘ splits factors into survival-active ($|\hat{\beta}| > 0.001$), genomics-only (PVE $> 1\%$), and dead. Running example: $K_{\text{init}} = 7 \to$ **2 survival-active** + **2 genomics-only** (4 kept, 3 pruned).

- **Main Takeaway:** Two stages, not one CV loop — CV-based C-index informs $K_{\text{init}}$ alongside three other criteria; ARD (not CV) determines $K_{\text{eff}}$.
- **Emphasis:** Say explicitly that cross-validation was *not* removed — it is one of four $K_{\text{init}}$ inputs. What changed is that it no longer single-handedly determines $K_{\text{eff}}$.
- **Likely Question:** "Are the ARD thresholds (PVE $> 1\%$, $|\hat{\beta}| > 0.001$) principled?" No. `beta_threshold` was calibrated to this model's own observed $|\hat{\beta}|$ scale (roughly 0.003–0.008 under YFB), not drawn from a citable sparse-factor-model source; `pve_threshold` is an uncited round number. A literature check and a sensitivity sweep are open items (ROADMAP.md) — flagged as a genuine limitation, not glossed over. See also the Limitations slide.

# Results

## Slide 12 — Section divider: Results

**Script.** (Divider — no content.) Signal the shift into results: simulation first, then real data, in that order.

## Slide 13 — Data: Synthetic Validation & PDAC Cohorts

**Script.** Two datasets, simulation first for the rest of Results. Simulation: EBMF-grounded ground truth, shared vs. study-specific factors, three scenarios, **two arms** (the projection model, EBMF $\to$ Cox) fit at an over-specified $K_{\text{init}}$ pruned by ARD, mirroring the real-data procedure. Real data: train TCGA-PAAD + CPTAC ($n \approx 273$), validate on five fully held-out cohorts, scored by the exact projection formula.

- **Main Takeaway:** The simulation now tests the *actual* K-selection procedure, not an oracle shortcut, and it comes before real data this time so the method is validated before it's applied.

## Slide 14 — Synthetic Results: over-specifying $K_{\text{init}}$ doesn't hurt performance

**Script.** $K_{\text{init}} = 6$ means the model was told the true number of factors directly, an idealized reference point never available on real data. $K_{\text{init}} = 12/20$ is the over-specify-plus-ARD-prune procedure actually used. C-index (and, in the appendix, recovery and specificity) are **flat across** $K_{\text{init}} = 6 \to 12 \to 20$ in every scenario, 15 seeds (up from 5, for a properly powered comparison).

- **Main Takeaway:** Over-specifying $K$ and trusting ARD costs nothing relative to knowing the true $K$ in advance, the core validation this simulation exists to provide.
- **Likely Question:** "Why only 2 arms now, not 5 like June?" $L\beta$ and the cohort-indicator arms are dropped from the whole narrative (see Slide 7); June already found no meaningful YFB-vs-YFB+cohort difference, so re-testing it here would add nothing new.

### Slide 15 — Synthetic Results: does a larger $K_{\text{init}}$ create more false alarms?

**Script.** The question: giving the model more factors than it needs gives ARD more non-signal factors it *could* mistakenly call survival-active. At 5 seeds it looked like a clean improvement; **at 15 seeds it softened** to a real but not-yet-significant trend: hybrid scenario $0.35 \to 0.22 \to 0.27$, paired $t$-test $K_{\text{init}} = 6$ vs. 12 gives $p = 0.07$. Negative-control scenario is flat at 0.10 regardless of $K_{\text{init}}$; the apparent 5-seed improvement there was noise.

- **Main Takeaway:** Report this as what it is, a promising trend, not a confirmed finding. Increasing to 15 seeds and watching the effect soften is itself worth mentioning as good practice, not something to skip past.
- **Good Line to Stress:** "A small, non-zero coefficient on a non-signal factor doesn't meaningfully move the risk score in practice, which is consistent with what we see here even when a false positive occurs."
- **Likely Question:** "Why show a non-significant result at all?" Because the direction is consistent and worth tracking, and stating the $p$-value plainly is more credible than either hiding it or overstating it as confirmed. It's also the direct follow-up to a concern raised at an earlier meeting about ARD over-counting survival-active factors.

### Slide 16 — Joint model vs. two-step as survival signal strength varies

**Script.** Reused from the 8/21 progress-book work. **"Strength" is a literal multiplier** on the true survival coefficients used to simulate outcomes, swept at 0, 0.25, 0.5, 1, and 2, with 1 being this project's standard realistic magnitude, so strength 2 is twice that. At strength 0 (no true signal at all), **joint $\approx$ EBMF-only**, no fabricated prognostic signal. At strength 2, the joint model pulls ahead. Only 10 replicate seeds, so no individual strength level is individually significant; the pattern across the whole sweep is the argument.

- **Main Takeaway:** Supervision is a free option: costs nothing when uninformative, helps when there is real signal.
- **Likely Question:** "What does 'signal strength 2' actually mean?" Twice the project's standard realistic-magnitude survival-coefficient setting, not an arbitrary or extreme value.

### Slide 17 — PDAC Cohorts: Prognostic Gene Programs

**Script.** Gene-weight heatmap, rows = programs, columns = genes (rotated from the progress-book report's tall version so it fits a slide), showing **only the 4 ARD-kept factors**. $K_{\text{init}} = 7 \to 2$ survival-active (Programs 3, 7) + 2 genomics-only (Programs 5, 6) kept. The 2 survival-active programs rest on **5 methods**: gene-set enrichment (fgsea and ORA), PurIST subtype correlation, hazard-ratio consistency across all 5 external cohorts, and DeSurv gene-list overlap. The 2 genomics-only programs (**desmoplastic stroma, immune infiltrate**) rest on **weaker evidence**: just 2 of those methods agreeing (enrichment and DeSurv overlap), not externally validated the way Programs 3/7 are.

- **Main Takeaway:** Supervision separates expression-only structure from prognostic structure. Be upfront that the genomics-only labels are preliminary, not equally validated.

- **Likely Question:** "How confident are we in 'desmoplastic stroma' and 'immune infiltrate' specifically?" A preliminary read, not a validated claim. The external-cohort check for these two programs was run as a negative control (confirming they carry no survival signal, which they shouldn't) rather than as positive evidence for the labels themselves. Testing whether the same two programs reappear on other cohorts is an open follow-up.
- **Likely Question:** "Why does the heatmap only show 4 factors, not all 7?" The 3 fully-pruned (dead) factors carry no real gene-program structure to show; ARD's own classification says so, and the pathway-enrichment work confirmed it (zero enriched gene sets for the dead factors).

### Slide 18 — PDAC Cohorts: Prognostic Association

**Script.** Unchanged from June. Split at median projection score $(YF)_k$: Program 7 adverse (MET, ITGA3, glycolytic genes), Program 3 protective (epithelial differentiation markers). Schoenfeld test: PH holds.

- **Likely Question:** "Does the joint $\beta$ sign always match the direction you report?" Not necessarily: Program 7's joint coefficient can run opposite its marginal adverse direction (suppression among correlated programs). We label by the empirical KM direction and stratify by $(YF)$ projection, not raw $\beta$ sign. Deliberate, not an inconsistency.

### Slide 19 — Choosing $K_{\text{init}}$: the consensus sweep

**Script.** The full $K_{\text{init}} = 2$–$20$ sweep. Each panel states its own preferred direction now: higher is better for ELBO, log-likelihood, and external C-index; lower is better for BIC. **ELBO and BIC both prefer $K_{\text{init}} = 3$** (both favor the smallest tested); BIC's complexity penalty (≈13,100 per unit $K_{\text{init}}$ here) dominates past single digits. **External C-index prefers $K_{\text{init}} = 12$**, though $K_{\text{init}} = 7$ through 20 mostly plateau around $C \approx 0.627$ (two isolated dips at $K_{\text{init}} = 11, 13$, a known single-seed CAVI factor-collapse artifact, not new). $K_{\text{init}} = 7$ kept as the recommendation: stable $K_{\text{eff}} = 2$, simplest single-fit path.

- **Main Takeaway:** The four criteria genuinely disagree; say so plainly, don't paper over it.
- **Good Line to Stress:** "I'm showing you the actual curves, not just the recommendation, because the disagreement itself is worth discussing."
- **Likely Question:** "Why not just use $K_{\text{init}} = 3$ if ELBO/BIC prefer it?" $K_{\text{init}} = 3$'s external C-index (0.594) is well below the $K_{\text{init}} \geq 7$ plateau (0.627); BIC's penalty doesn't know about held-out generalization, only in-sample fit complexity. We weight external validity heavily, which is why $K_{\text{init}} = 7$, not 3, is the practical recommendation.
- **Likely Question:** "Why not $K_{\text{init}} = 12$, since external C prefers it?" A legitimate alternative; $K_{\text{init}} = 7$ through 20 are close on the plateau. $K_{\text{init}} = 7$ is kept for simplicity (single fresh fit) and because $K_{\text{eff}}$ is identical (2) across that whole range, so nothing about the biological conclusion changes with the choice.

### Slide 20 — Joint model vs. the unsupervised two-step

**Script.** Real-data analog of the synthetic comparison. Two arms: the projection model vs. the **fairest** independent two-step baseline (EBMF → Cox, $K = 40$, LASSO stage 2: large, YFB-uninformed $K$, regularized stage 2, not a small $K$ or an overfitting plain `coxph()`). Projection mean C = **0.627** vs. EBMF **0.601**; pooled bootstrap difference **+0.026, 95% CI [0.0002, 0.0498], significant** ($n = 616$ pooled).

- **Main Takeaway:** This is the hardest-to-argue-with two-step comparison available, not a favorable cherry-pick.
- **Emphasis:** This CI was independently re-derived from paired risk scores before this talk, not quoted from memory; say so if pressed on where the number comes from.
- **Likely Question:** "Why $K = 40$ for the baseline, not something smaller?" Larger $K$ gives the unsupervised step more capacity to find whatever structure exists, without ever seeing survival outcomes. Handing it a *smaller* $K$ (like 7, matched to YFB) actually hands over part of the answer, since the joint model chose that $K$ using its own selection procedure. $K = 40$ with a regularized (LASSO) stage-2 Cox is the configuration built specifically to be the fairest test, not the one most favorable to us.

### Slide 21 — Model Comparison Summary

**Script.** Two rows: the projection model vs. the unsupervised two-step. Table note explains $K_{\text{init}}$ (consensus-selected) vs. $K_{\text{eff}}$ (ARD-determined) explicitly. Note the two-step baseline's $K_{\text{eff}}$ is no longer "n/a": it's **15 of the 40 EBMF factors**, the count that keep a nonzero coefficient after LASSO regularization, a different mechanism from ARD (L1 shrinkage vs. an ARD prior on $\beta$) but answering the same question.

- **Main Takeaway:** Recommendation unchanged in substance (projection predictor, per-platform z-std, no cohort indicator); the row count and $K$-selection description are what changed, plus the two-step baseline now has a real $K_{\text{eff}}$ instead of a placeholder.

### Slide 22 — Impact / key findings

**Script.** Five points: joint beats two-step at no cost when uninformative; preprocessing is decisive; $K$ selection is the two-stage split (say it exactly the same way as Slide 11: CV informs $K_{\text{init}}$, ARD determines $K_{\text{eff}}$); over-specifying $K_{\text{init}}$ costs nothing (simulation, 15 seeds); mean external C = **0.627**. Then limitations, stated plainly: it isn't obvious which $K_{\text{init}}$ is best; the ARD thresholds aren't literature-derived, and a fixed threshold also assumes every factor's projection score is on a comparable scale, which is only partly true (see below); small cohorts are noisy; biological annotation is strong for the two survival-active programs, preliminary for the two genomics-only ones.

- **Main Takeaway:** Every limitation named here has already been raised on the slide that produced it; this is a recap, not new information, which is the point: nothing saved for later or hidden.
- **Likely Question (raised at the last advisor meeting):** "Are the $\beta$ coefficients even comparable across factors, so a single threshold can be applied to all of them?" Partially. Each factor's loading vector is unit-normalized before the projection score is computed, consistently at train and test time, which removes the dominant scale gap (an unnormalized loading on more genes would otherwise get a mechanically larger projection score for reasons unrelated to survival relevance). That does *not* guarantee every factor's projection score has the same *variance* across patients, which still depends on which genes each factor loads on. This normalization was built for fitting stability, not explicitly

to solve comparability, so it was never checked whether it's sufficient. Tracked in ROADMAP.md as a next step: test whether $\mathrm{Var}(ZF_k)$ actually varies across the kept factors, and if so, whether standardizing it changes which factors clear the threshold. Not a quick fix: it would change the model's fitting itself, not just a display choice, so it needs its own dedicated investigation, not a same-day patch.

### Slide 23 — Future directions & summary

**Script.** Current status, not "addressed since June" framing. Next steps: lead with **multiple cohorts / multiple modalities via indicator columns in $L$**: stack methylation + expression, possibly modality-specific priors, leave-one-study-out and train-on-A-vs-B-vs-A+B validation. Then manuscript prep. Summary panel restates the recommended configuration with numbers.

- **Main Takeaway:** The cohort-indicator idea from June is repositioned here as a forward-looking direction (multi-cohort/multi-modality), not a tested-and-rejected result; don't imply it was tried and failed.

### Slide 24 — Discussion

**Script.** Open the floor. Appendix covers the full $K_{\text{init}}$ sweep table, simulation detail (recovery and specificity), convergence diagnostics, and the DeSurv comparison.

## Anticipate These Audience Questions

- **"Didn't you say cross-validation was replaced by ARD? Isn't that inconsistent with C-index still being on the K-sweep slide?"** No inconsistency: cross-validated external C-index is one of four inputs that help choose $K_{\text{init}}$ (Stage 1). What ARD replaced is using cross-validation to determine $K_{\text{eff}}$ (Stage 2) directly; that used to require a second validation loop, and now it falls out of a single over-specified fit.
- **"Are the $\beta$ coefficients on a comparable scale across factors, so a single threshold makes sense?"** See Slide 22's note above: partially. Loading vectors are unit-normalized, which removes the dominant scale gap, but that doesn't equalize projection-score variance across factors, and nobody has checked whether that residual gap matters. Tracked as an open item, not resolved today, and not a quick fix since it touches the model's fitting, not just a display choice.
- **"The four $K_{\text{init}}$ criteria disagree by a lot, doesn't that undermine the method?"** It's a real finding, not a flaw to hide: BIC penalizes model complexity in a way that doesn't know about held-out generalization, so it favors small $K_{\text{init}}$; external C-index does know about generalization and prefers more capacity. The biological conclusion ($K_{\text{eff}} = 2$ survival-active) is stable across the whole $K_{\text{init}} = 7$–$20$ plateau regardless of which criterion wins, which is the more important robustness check.
- **"Why did the two-step comparison number change from what I remember hearing before (0.581 or similar)?"** That was an earlier, less carefully regularized baseline (smaller $K$, plain `coxph()` that overfits at large $K$). The current 0.601 baseline uses $K = 40$ with a LASSO-regularized stage-2 Cox specifically to remove that overfitting artifact and give the unsupervised approach every reasonable advantage before comparing.

- **"Why present a non-significant result (the false-positive-rate trend) at all?"** Because the direction is consistent across 15 seeds even though the test isn't significant, and stating the $p$-value plainly is more credible than either omitting the result or overstating it as confirmed. It's also directly relevant to a concern raised in an earlier meeting about ARD over-counting survival-active factors; this is the follow-up data on that concern.
- **"How does this compare to the lab's penalized supervised-MF (DeSurv) method?"** Bayesian counterpart: same supervised NMF + Cox idea, same TCGA + CPTAC training and five external cohorts, same projection-based scoring, borrowed combined-rank gene selection. Differences: DeSurv tunes $k, \alpha, \lambda$ jointly by Bayesian optimization; this model selects $K_{\text{init}}$ by the four-criteria consensus and lets ARD prune, with empirical-Bayes priors learning shrinkage (no $\alpha, \lambda$ to tune) and posterior uncertainty. Different primary metrics reported (pooled HR/SD vs. mean external C), so a matched same-protocol head-to-head remains a planned next step.
- **"What happened to the cohort-membership indicator from the June deck?"** Not dropped as a failed idea; repositioned as a forward-looking multi-cohort/multi-modality direction (Future Directions), since the more useful framing now is extending $L$ to combine multiple cohorts and/or modalities (e.g. methylation + expression), not re-litigating June's single-modality result.

## Appendix Slide Scripts

### Slide A1 — Appendix: $K_{\text{init}}$ sweep, full table

**Script.** If someone wants the actual numbers behind the sweep curves, this is the table: all 19 $K_{\text{init}}$ values with survival-active/genomics-only/dead counts, ELBO, BIC, log-likelihood, and external C, $K_{\text{init}} = 7$ highlighted.

- **Main Takeaway:** Full transparency on the sweep; nothing on the curve slide is cherry-picked.

### Slide A2 — Appendix: simulation detail, factor recovery & specificity

**Script.** Backup for Slide 14, in case someone wants to see recovery and specificity, not just C-index. Same flat-across-$K_{\text{init}}$ story in every scenario.

- **Main Takeaway:** The C-index slide's null result isn't cherry-picked either; every metric tells the same flat story.

### Slide A3 — Appendix: Convergence & survival-activity magnitudes

**Script.** Unchanged from June. RMSE stabilizes to a plateau while the ELBO, the actual objective, governs stopping (relative change below $10^{-5}$). The $|\hat{\beta}_k|$ bars show the shrinkage: most programs fall below threshold, a couple clear it.

- **Main Takeaway:** Convergence is judged on the ELBO; $K_{\text{eff}}$ is read off the few programs with $|\hat{\beta}|$ above threshold.

### Slide A4 — Appendix: Relation to DeSurv

**Script.** Backup for the inevitable comparison question (see also the Anticipated Questions section above). Be collegial and precise: Bayesian sibling of DeSurv, same data and scoring, genuine differences are adaptive priors and fewer tuned knobs, plus posterior uncertainty. Different primary metrics reported; a matched head-to-head is the right next step, not a superiority claim today.

- **Main Takeaway:** Same problem and data, Bayesian estimation; head-to-head is planned, not done.